



Red Hat One

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Global | 2026



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WHAT IS THIS SESSION?

- **Purpose** - This breakout session is for **all audiences**, aimed at the **intermediate technical** level - designed to equip SSPs, SAs, Consultants and Consulting Architects with everything they need to confidently position and discuss Ansible development workspaces which uses Dev Spaces on Red Hat OpenShift for AAP automation content development.
- **Target Audience**
 - Primary: Specialist Solutions Architects (SSAs) and Consultants
 - Secondary: Solutions Architects (SAs) and Consulting Architects
 - Technical Level: 2/2 (sets up customer challenges but dives into advanced features)
- **What Makes This Different**
 - This is **FOR sellers** and **implementers**, not FOR customers (though some slides may be reusable)
 - Focus on **talk tracks**, not feature lists
- **Session Goals**(By the end of this session, sellers will have):
 1. Challenges unique to AAP customers
 2. Various solutions we have for these different challenges
 3. Introducing Tech Preview for Ansible development workspaces
 4. Art of the possible when using OpenShift not just for Dev Spaces but running AAP and many additional capabilities
 5. Resources and next steps

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Developing Ansible automation content in a browser-based environment

BO1507

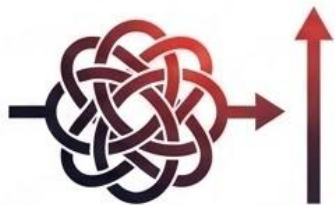
Vinny Valdez
Chief Architect – Automation
NA Field CTO Org

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Sr Technical Marketing
Manager
Ansible BU

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Automation TDP
NAPS

Jeff Pullen
Principal Architect
NAPS

What you'll be able to walk away with after this session...



Major Challenges & Solutions

Understanding the hurdles in automation content development and the solutions we offer.



One-Click Dev Environment

Leveraging Ansible development workspaces to integrate all necessary tools seamlessly.



Art of the Possible

Unlocking new potential and exploring advanced capabilities in automation.

Automation Pipeline

**Developing Ansible
automation content in a
browser-based
environment
Wednesday Jan 28 8:30
AM PST**

Speed up Ansible
Automation Hub collection
release using CI/CD strategy
Wednesday Jan 28 2:15 PM

Ac

An End-to-End Ansible
Lifecycle in Action
Wednesday 28 9:45 AM PST

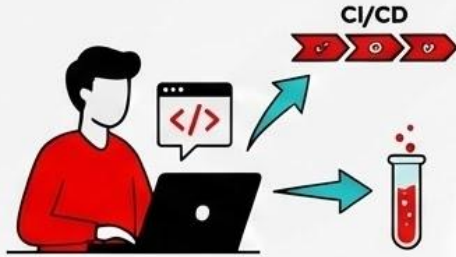
* In the repository this Software Bill of Materials is referred to as a Manifest



What you'll be able to walk away with after this session...

- Major challenges involved in proper automation content development and our solutions
- Explore how Ansible development workspaces leverages all of these tools in a one-click dev environment
- Art of the possible

Challenges



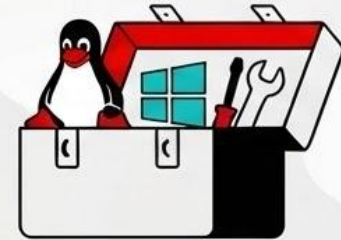
Software development as an Automation Engineer

- Automation not always treated as code
- git and CI/CD proficiency required
- Little if any testing



Challenging On-boarding & inconsistent environments

- Multi-page manual dev setup
- Multiple steps and approvals
- “Works on my machine” fallacy



Capabilities to develop Ansible properly

- Requires Linux
- Using Windows for development complex
- Need to gather and manage dev tools



Challenges

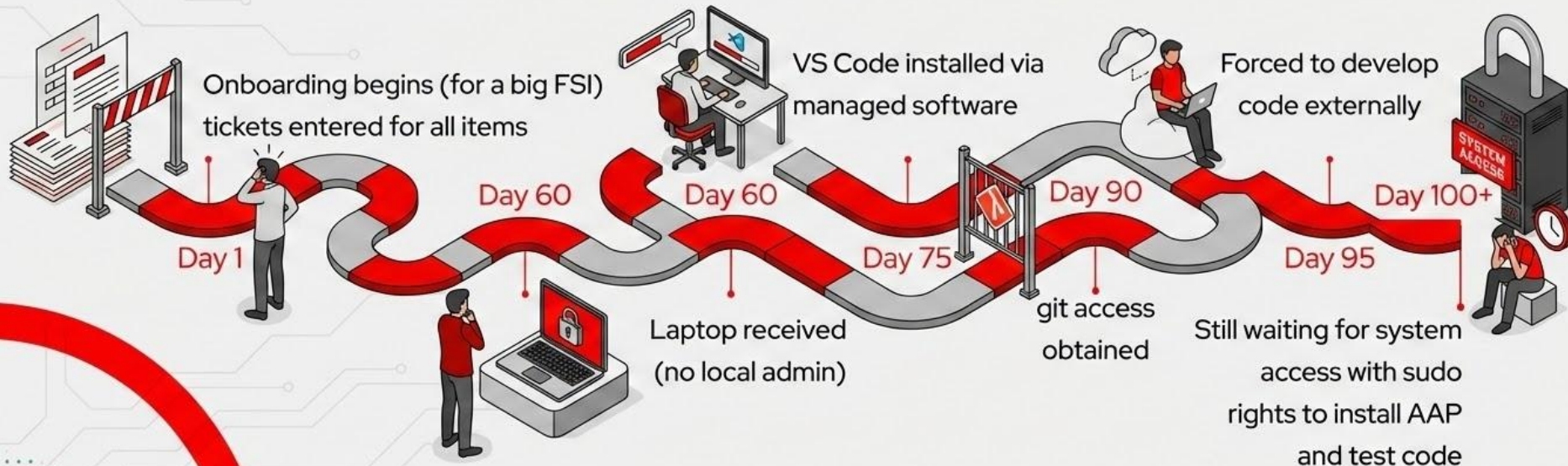
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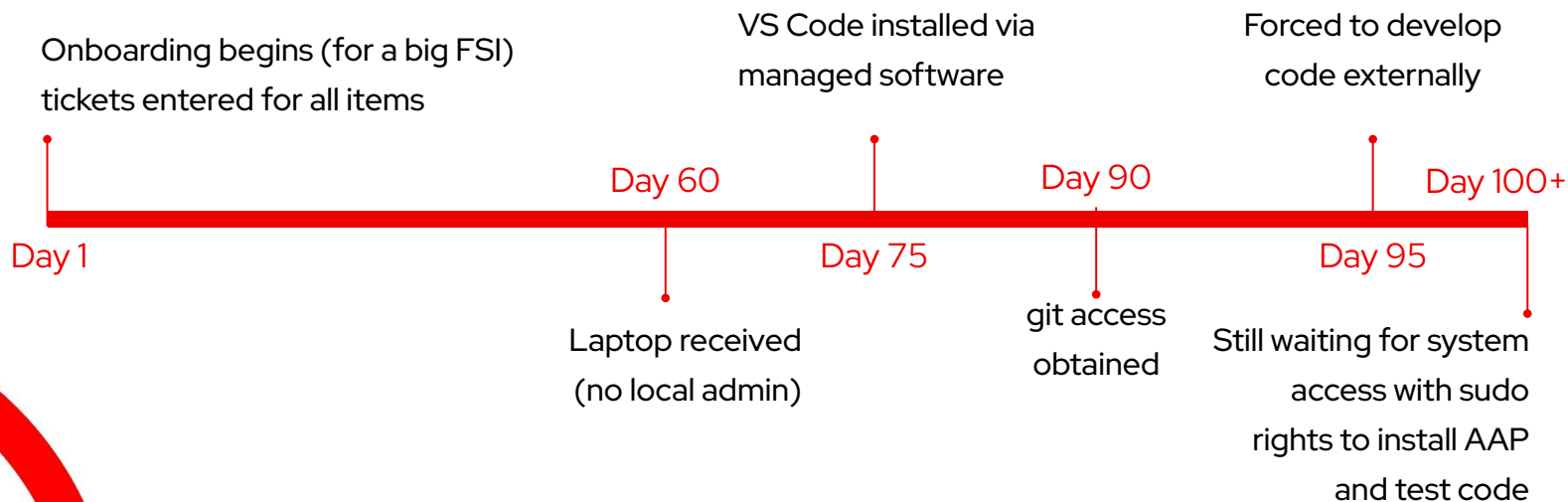
Automation Engineers are Developers, like it or not



Even if you have git and CI/CD proficiency, slow onboarding can be a drag



Even if you have git and CI/CD proficiency, slow onboarding can be a drag







Ansible Development Tools

Amazing tools need somewhere to run



Ansible development tools (ansible-dev-tools)

Supported, enterprise-grade components to enable creating, testing, and deployment of Ansible

Create

ansible-creator

Scaffold projects and content

ansible-navigator

Develop and troubleshoot with execution environments

Ansible development environment (ade)

Build and manage virtual environments for content development

Test

Ansible Molecule

Functional test runner for collections, playbooks and roles

ansible-lint

Identify and correct stylistic errors and anti-patterns

pytest-ansible

Added functions for module and plugin Python code unit testing

tox-ansible

Test collections under different Python interpreters and Ansible core versions

Deploy

ansible-builder

Build Ansible execution environments

ansible-sign

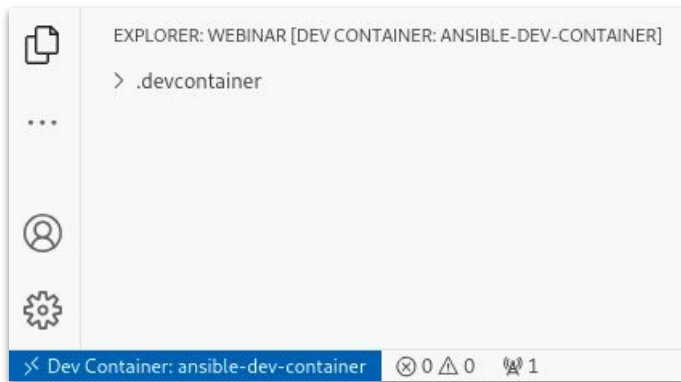
Sign and verify Ansible content



Ansible VS Code extension and Dev Container

Normalize and Isolate Development Environments

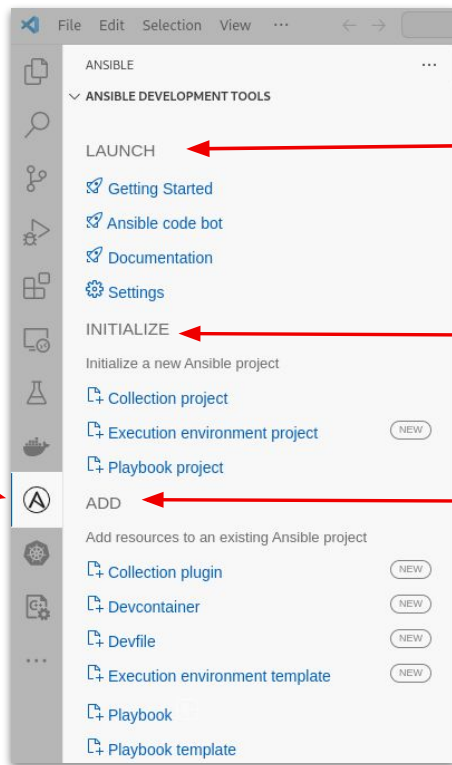
- ▶ Create containerized development environments with all Ansible development tools included
- ▶ Compatibility with macOS, Windows and Linux
- ▶ Access Microsoft Dev Containers extension through VS Code marketplace



Active dev container
in VS Code

Ansible VS Code extension

Integrating tools into a unified UI



Access the Ansible extension

Learn more about the extension and Ansible development tools

Initialize projects with ansible-creator

Generate playbooks, add project resources, and build EEs

A screenshot of the 'Create new Ansible collection' form. The form has a title 'Create new Ansible collection' and a subtitle 'Streamlining automation'. It contains fields for 'Namespace' (acme), 'Collection' (acme_tools), and 'Init path' (/root/.ansible/collections/ansible_collections). Below the fields, it shows the 'Collection name' (acme.acme_tools) and the 'Project path' (/root/.ansible/collections/ansible_collections/acme/acme_tools). Red arrows point from the text labels to the form fields.

Forms for creating content via the VS Code UI



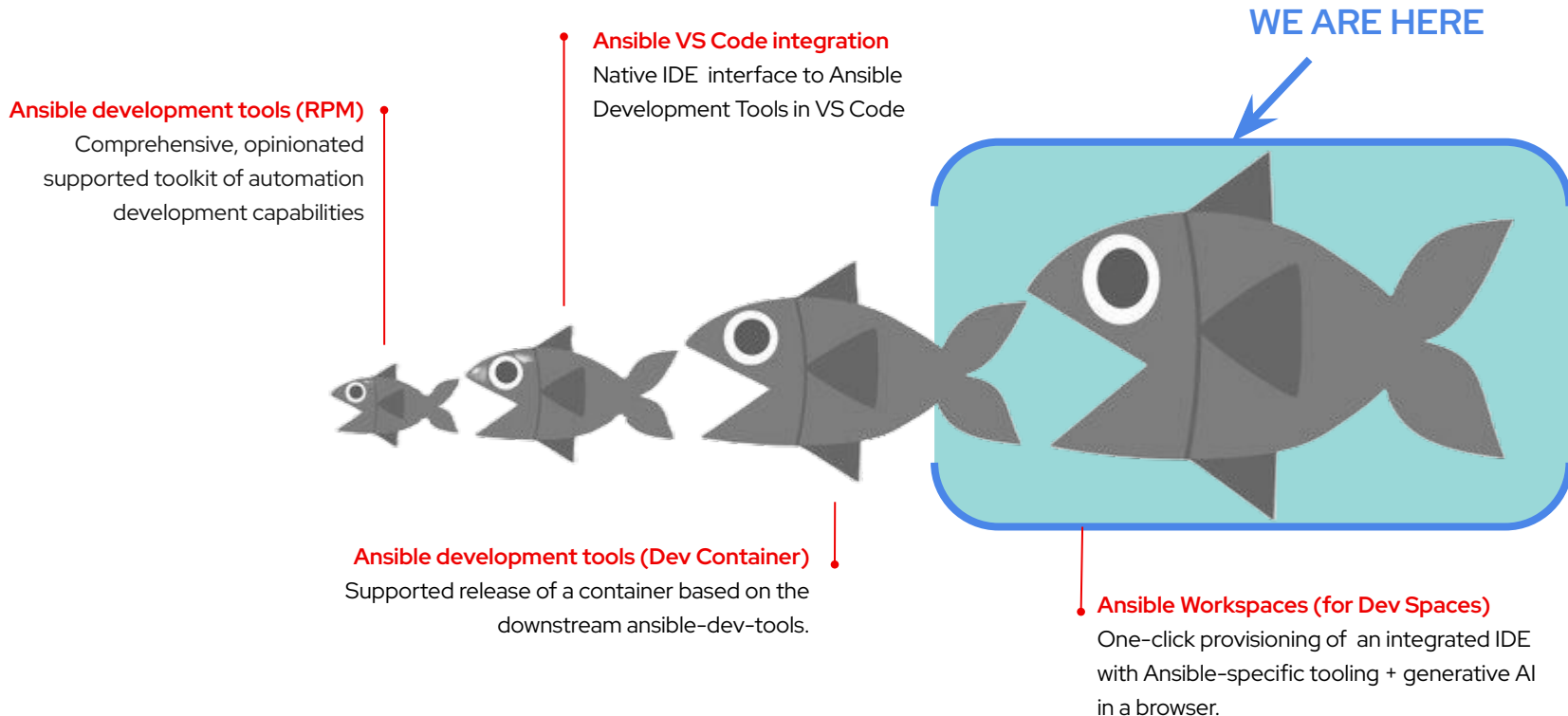
Ansible Development Workspaces

Bringing it all together



The Ansible development tools experience

CONFIDENTIAL



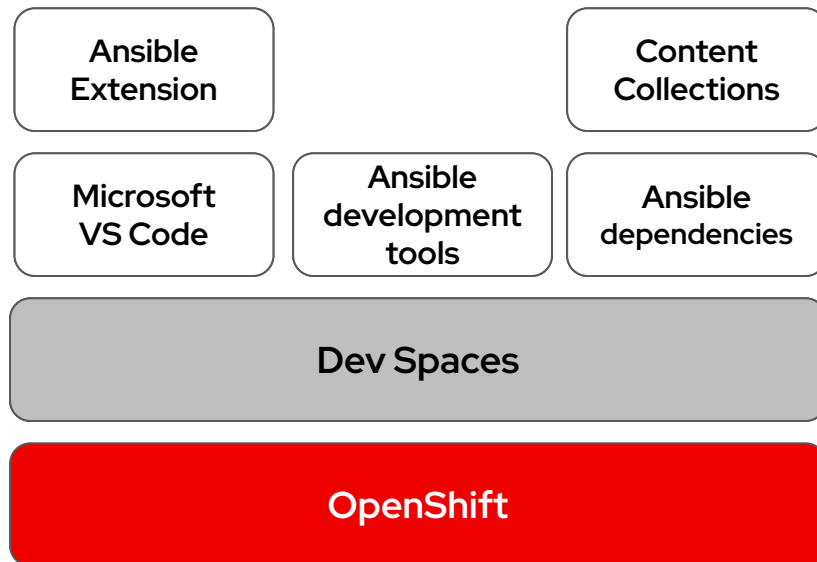
Ansible development workspaces

What are Ansible development workspaces?

Enterprise-ready on-demand Ansible content development environments

What is it?

- ▶ One-click provisioning of standardized, secure, environments based on Ansible best practices
- ▶ Everything automation developers need in a browser – VS Code, Ansible development tools, Ansible Lightspeed, documentation, etc.
- ▶ Supports Ansible content development on Windows and Mac OS workstations
- ▶ Enables IT organizations centralized management and policy enforcement



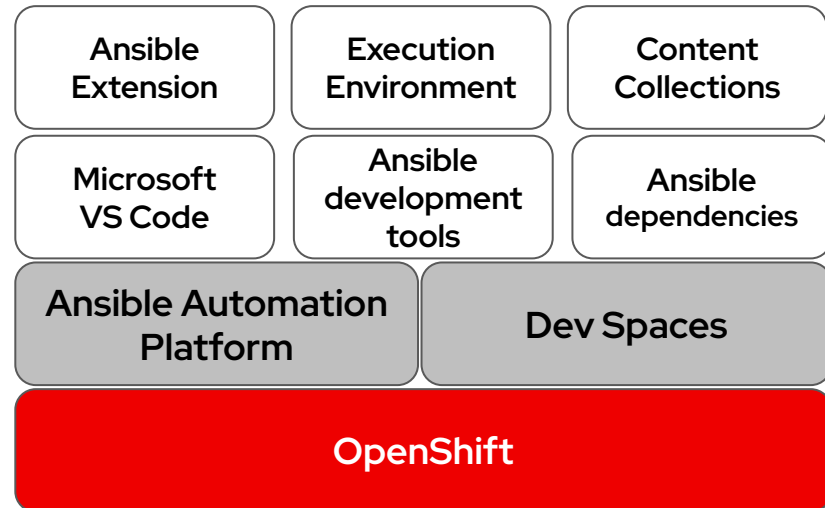
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OpenShift Dev Spaces

Red Hat OpenShift Dev Spaces uses Kubernetes and containers to provide developers and other IT team members with a consistent, secure, and zero-configuration development environment. The experience is as fast and familiar as an integrated development environment (IDE) on your laptop.

**Developer
Workstation**

**OpenShift
(Kubernetes)**

**Dev Spaces
Workspace**

Git Repositories

CI/CD Pipelines

Ansible development workspaces

Within Red Hat OpenShift, launch Dev Spaces

The screenshot shows the Red Hat OpenShift console interface. On the left is a sidebar with navigation links: Home, Favorites, Ecosystem, Helm, Workloads, Networking, Storage, Builds, Pipelines, Observe, Compute, User Management, and Administration. The main content area is titled 'Overview' and shows 'Cluster' as the selected tab. It includes sections for 'Getting started resources' (with links to 'Set up your cluster', 'Take console tour', and 'Configure alert receivers'), 'Build with guided documentation' (with links to 'Enable the Developer Perspective' and 'Impersonating the system:admin user'), and 'View all steps in documentation' and 'View all quick starts'. A dropdown menu is open on the right, showing options: 'Red Hat Applications' (with sub-items 'OpenShift Cluster Manager' and 'Red Hat Hybrid Cloud Console'), 'Red Hat OpenShift Dev Spaces' (highlighted with a blue link icon), 'OpenShift GitOps' (with sub-item 'Cluster Argo CD'), and 'OpenShift Lightspeed' (with sub-item 'Your personal AI helper'). Below the dropdown, there are links to 'See what's new in OpenShift'. At the bottom, there are three panels: 'Details' (showing 'Cluster API address' and 'Cluster ID'), 'Status' (showing 'Cluster', 'Control Plane', 'Operators', 'Dynamic Plugins', and 'Storage' with green checkmarks), and 'Activity' (showing 'Ongoing' and 'Recent events' with a 'Pause' button).

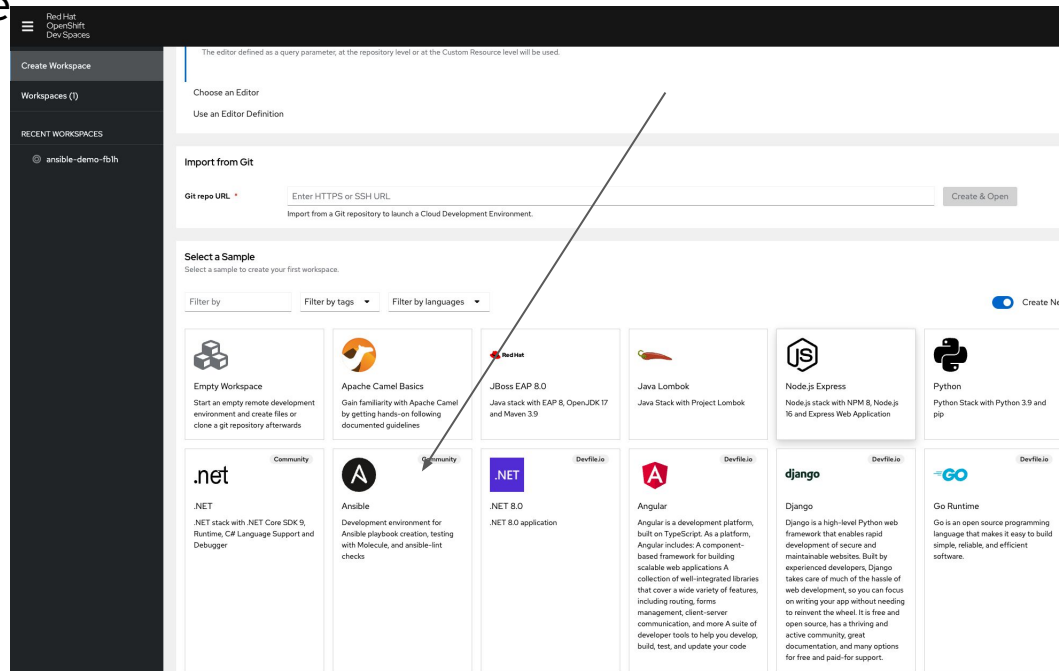


In Dev Spaces, launch Ansible workspace

launch provided creator image

Sample is provided

- ▶ Ansible creator image
- ▶ Contains ansible development tools
- ▶ Batteries included



Ansible development workspaces

A workspace spins up according to the devfile

Workspaces > Starting workspace ansible-demo-e6nf

Starting workspace ansible-demo-e6nf Starting

Progress Logs Events

- ✓ 1 Initializing
- ✓ 2 Checking for the limit of running workspaces
- ✓ 3 Creating a workspace
- 4 **Waiting for workspace to start**
 - ✓ DevWorkspace is starting
 - ✓ Resolved plugins and parents from DevWorkspace
 - ✓ Storage ready
 - ✓ Networking ready
 - ✓ DevWorkspace serviceaccount ready
 - ✓ DevWorkspace secrets ready

Waiting for workspace deployment

⌚ Downloading IDE binaries... (it can take a few minutes)

5 Open IDE



Ansible development workspaces

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Developer Sandbox

EXPLORER

(WORKSPACE)

- ansible-devspaces-demo
 - .vscode
 - collections
 - .ansible-lint
 - .code-workspace
 - .pre-commit-config.yaml
 - .yamllint
 - devfile.yaml
 - README.md
 - syntax_highlighting_demo.yml

devfile.yaml

```
ansible-devspaces-demo > ! devfile.yaml > ...
Devfile - Devfiles (devfile.json)
1 # cspell: disable=devspaces
2 schemaVersion: 2.2.2
3 metadata:
4   name: ansible-demo
5 components:
6   - name: tooling-container
7     container:
8       image: ghcr.io/ansible/ansible-devspaces@sha256:ce1ecc3b3c350eab2a9a417ce14a33f4b222a6aafd663b5cf9
9       memoryRequest: 256M
10      memoryLimit: 6Gi
11      cpuRequest: 250m
12      cpuLimit: 2000m
13      args: ['tail', '-f', '/dev/null']
14      env:
15        - name: 'ANSIBLE_COLLECTIONS_PATH'
16          value: '~/ansible/collections:/usr/share/ansible/collections:/projects/ansible-devspaces-demo
17        - name: 'KUBEDOCK_ENABLED'
18          value: "true"
19      commands:
20
21      - id: molecule-create
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL ANSIBLE OPENSIFT TERMINAL

ansible-devspaces-demo \$ adt --version

ansible-builder	3.1.0
ansible-core	2.18.7
ansible-creator	25.8.0
ansible-dev-environment	25.8.0
ansible-dev-tools	25.8.3
ansible-lint	25.8.2
ansible-navigator	25.8.0
ansible-sign	0.1.2
molecule	25.7.0
pytest-ansible	25.8.0
tox-ansible	25.8.0

ansible-devspaces-demo \$

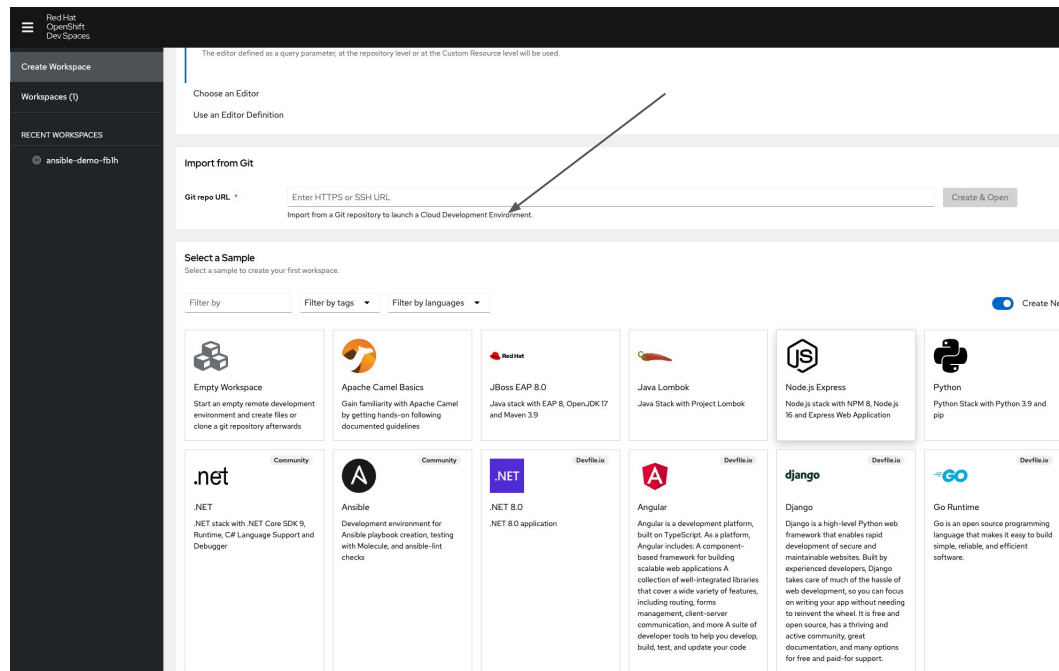
Ln 1, Col 1 Spaces: 2 UTF-8 LF YAML Layout: U.S. Devfile(2.2.0)



Dev Spaces for Power Users Advanced Capabilities

Custom Dashboard

- ▶ Samples are configurable
- ▶ Batteries included
- ▶ Custom tasks can be included
- ▶ Additional permissions and/or resources configured



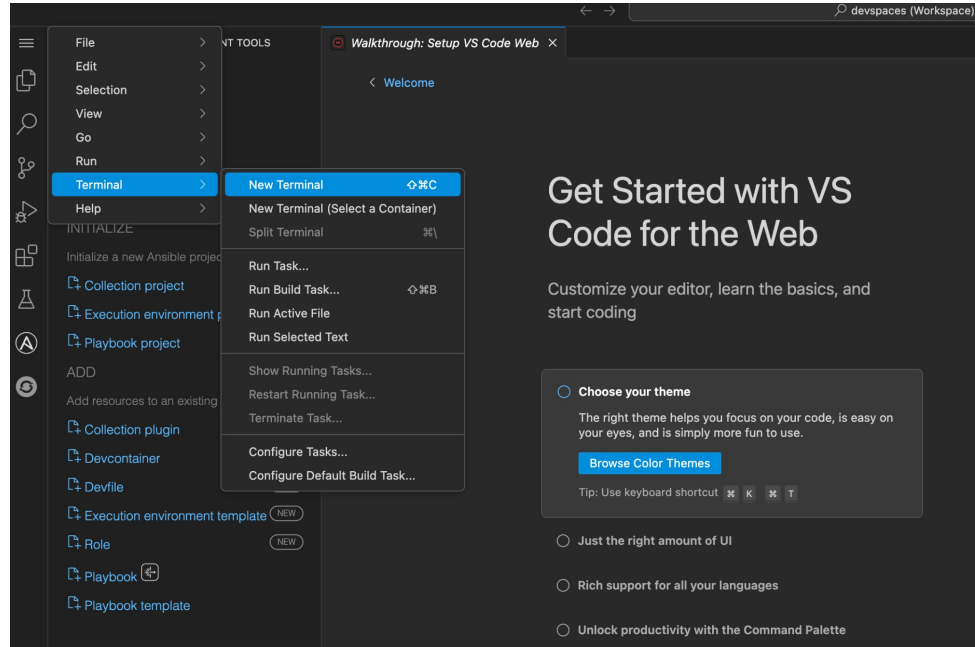
Ansible development workspaces

Dev Spaces for Power Users

Advanced Capabilities

Fully Functional Terminal

- ▶ No need to install things locally
 `"command not found: ansible"`
- ▶ Downstream API inherits OCP permissions
 `"oc whoami"`



Dev Spaces for Power Users

Advanced Capabilities

Integration with Source Control

- ▶ Automatically setup credentials for Gitlab, Github, Bitbucket and Azure
- ▶ Leverages OAuth integration to reduce dependency on static secrets installed on laptops/servers

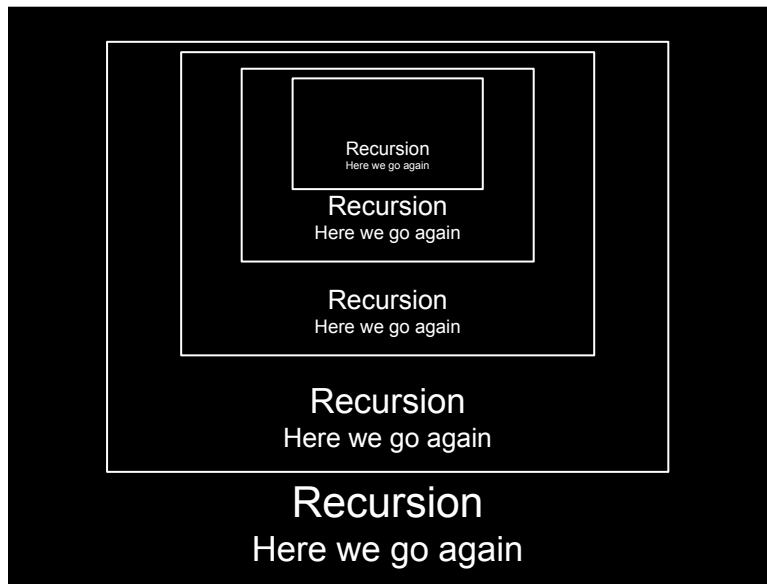


Dev Spaces for Power Users

Advanced Capabilities

Linux User Namespaces

- ▶ GA in OCP 4.20
- ▶ **Nested containers** now supported
- ▶ Ansible-navigator using EEs - no more KUBEDOCK
- ▶ Ansible-builder, podman, buildah



Art of the Possible



**A Trojan horse
for OpenShift**



**A reason to buy
OCP Virtualization**



**Dev Spaces is more
than just an IDE**

- pre-coded script logic (tasks)
- manage credentials (PATs)
- scenarios



Art of the Possible

- A Trojan horse for OpenShift
- A reason to buy OCP Virtualization
- Dev Spaces is more than just an IDE
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What else does OpenShift bring for AAP?

Feature	Traditional VM-Based AAP	AAP on OpenShift
Scaling	Fixed capacity (pay for peak usage)	Dynamic scaling (pay for what you use)
Upgrades	Manual, risky, requires downtime	Automated via Operator
Developer Experience	Inconsistent local laptops	Standardized Dev Spaces
Resilience	High Availability is complex to setup	Native HA & Self-healing



Customer Profile

US Government Financial Institution

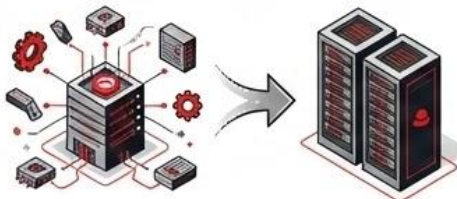
Challenges:

- Hyperfragmented organization
- Mixed experience



Goals:

- Consolidate redundant services
- Migrate to a new datacenter



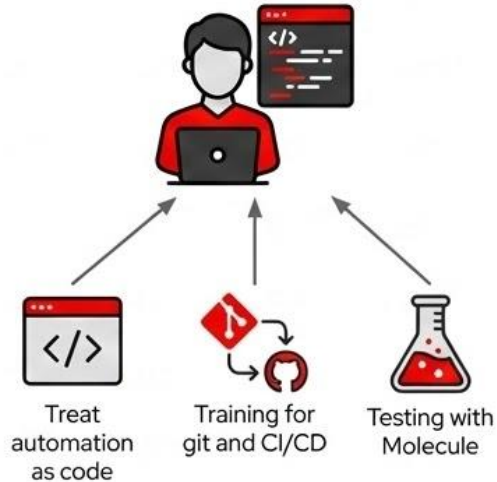
The Method:

- Piggyback on other opportunities
- Lead by example

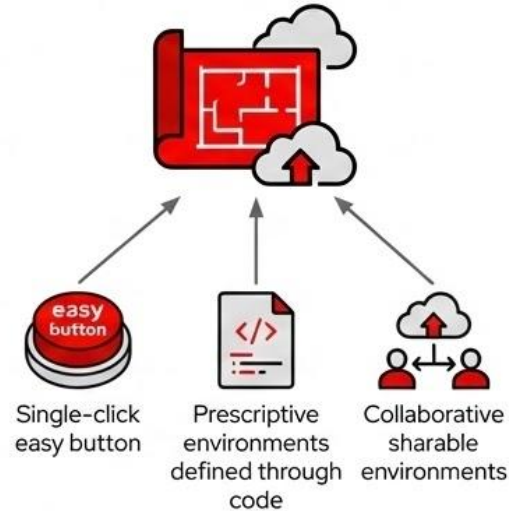


How Ansible Development Workspaces helps addresses these Challenges

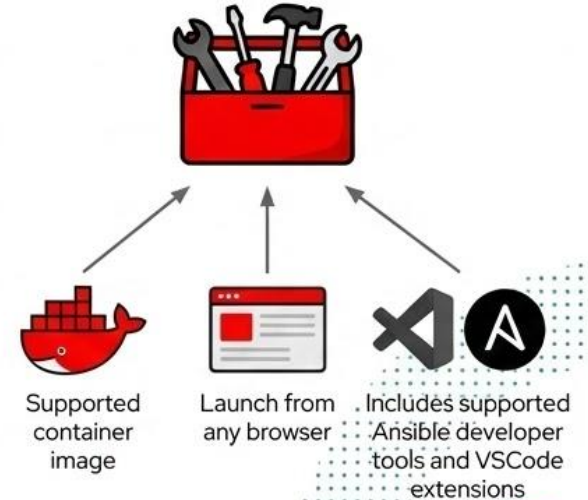
Software development as an Automation Engineer



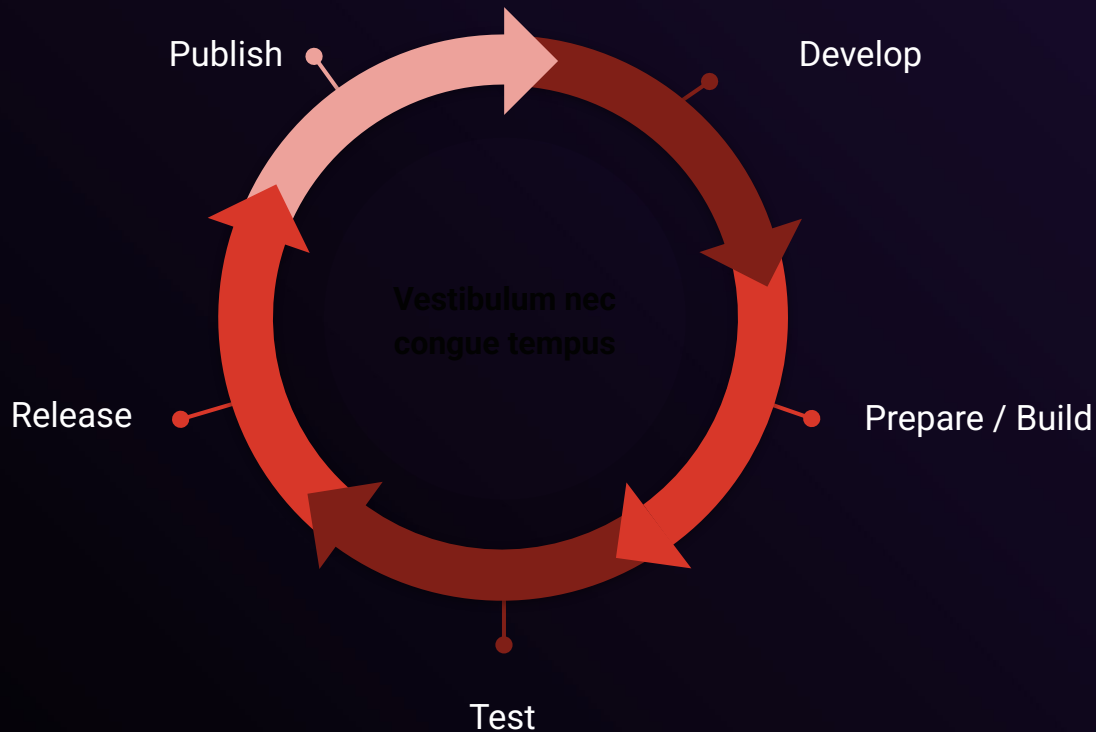
Challenging On-boarding & consistent environments



Capabilities to develop Ansible properly



Collection Pipeline Overview



How Ansible Development Workspaces helps addresses these Challenges

- Software development as an Automation Engineer
 - Treat automation as code
 - Training for git and CI/CD
 - Testing with Molecule
- Challenging On-boarding & consistent environments
 - Single-click easy button
 - Prescriptive environments defined through code
 - Collaborative sharable environments
- Capabilities to develop Ansible properly
 - Supported container image
 - Launch from any browser
 - Includes supported Ansible developer tools and VSCode extensions



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Resources



Getting Started

And more

- Read the documentation - [Using Ansible development workspaces for automation content development](#)
- [Deploy Dev Spaces on OpenShift](#)
- [Advanced Ansible Bootcamp on RHDP](#)
- Ansible Automation Superlab: The Enterprise Edition (coming soon)



More Information

- Differences and workarounds in OpenShift 4.20
- Support for deploying pods and containers into Linux user namespaces is now generally available and enabled in OpenShift 4.20



Thank you

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When Do I need OpenShift?

- When WSL is totally locked down via corporate policies
- Certain features of AAP 2.6 are currently only available on OpenShift (Self-Service Automation Dashboard)
- Managing several deployments of AAP - OpenShift provides an Operator to handle the entire lifecycle of all AAP components
- Not just useful for AAP - OpenShift offers a massive leap in operational maturity, scalability, and developer experience that is difficult to achieve with standalone VMs
- Dynamic Execution Capacity through Container Groups
- Self-Healing Infrastructure



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